MECHANICS OF SOLIDS PROJECT 2026–27 · TEAM "METTAMATERIALS"

GEOMETRY THAT PUSHES BACK

A review of four architected lattices — what they do, how well the models predict them, and what they are good for.

Repository — code, data, full reports and CAD:
github.com/ErrorNik14/metamaterials-and-applications

Squeeze a block of ordinary material and it bulges sideways. The lattices in this project are built to misbehave: three are *auxetic* — compress them and they pull inward — and the fourth replaces solid connections with struts and cables held taut by self-stress. In every case the behaviour comes from geometry, not from the material it is printed in, so it can be designed. Each strand follows the same arc: a mechanism, closed-form models, a finite-element check in ANSYS, and an application.

01 · RE-ENTRANT

Bow-tie honeycomb

Four models vs ANSYS; a running-shoe midsole

ν = −0.14 forefoot cell, ANSYS

02 · CHIRAL

Hexachiral lattice

Rings and winding ligaments; a morphing airfoil

ν = −0.82 3D coupon, ANSYS

03 · ROTATING SQUARES

Counter-rotating tiles

Rigid-unit theory, own FE, a TPU specimen

ν = −0.70 TPU specimen, ANSYS

04 · TENSEGRITY

Struts and cables

Graded hexagonal cells; 2D sheets to 3D columns

0.30 mm max deflection, ANSYS

STRAND 01

Re-entrant

Bow-tie cells that fold shut — from four models to a running-shoe midsole

Turn the inclined walls of a hexagonal honeycomb inward and every cell becomes a bow-tie. Under compression the bow-ties fold shut, drawing material *toward* the load instead of pushing it out — the opposite of a conventional honeycomb, which barrels (Fig. 1). Four models describe the cell, each adding a mechanism: Gibson & Ashby (bending), Masters & Evans (plus stretching), Malek & Gibson (thick walls, rigid joints, shear) and Ohtaki (large rotations). The linear three share one compliance form:

$$E_2 = \frac{E_s\,\beta^3(\alpha + \sin\theta)}{\cos\theta\,\left(\cos^2\theta\,C_p + \sin^2\theta\,C_a + C_v\right)}, \qquad \alpha = h/l,\ \ \beta = t/l$$

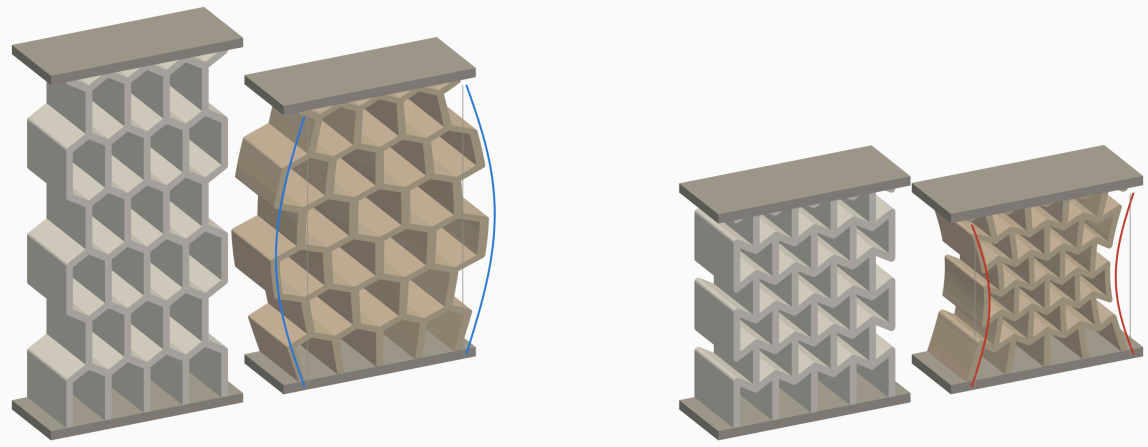


Fig. 1 Same compression, opposite response: a hexagonal block barrels out (left), a re-entrant block draws in (right).

Models against ANSYS

A printable cell (θ −30°, t/l = 0.4) was simulated as a 4 × 4 plane-stress block, then a batch of 25 cells ran through ANSYS APDL. Every model over-predicts the auxetic effect — thick printed joints resist the folding — while Malek & Gibson, the only model with rigid joints, tracks the stiffness best.

±10 %
Malek & Gibson stiffness error, all 25 cells

−0.14
ANSYS ν_{21}, forefoot cell (models −0.44 to −1)

0.36 MPa
buckling stress; no model predicts it

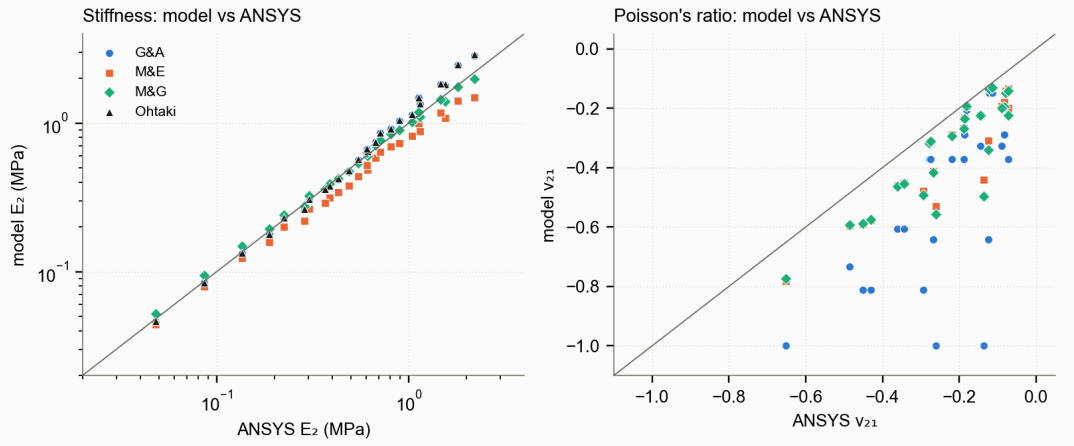


Fig. 2 Four models against ANSYS for 25 cells: stiffness (left) and Poisson's ratio (right).

A midsole under a running foot

A 270 mm midsole was filled with 34 lattice slices and loaded by a measured stance (Fukuchi dataset, 213 ms, peak 1523 N) through a three-segment pseudo-foot posed frame by frame. Each slice is a beam on a bed of lattice columns; the foot sinks until the slices return the measured force. A grid search over 24 cells, inspired by swarm-optimised midsole design, mapped how far the foot sinks (*dip*) against how much energy the sole absorbs.

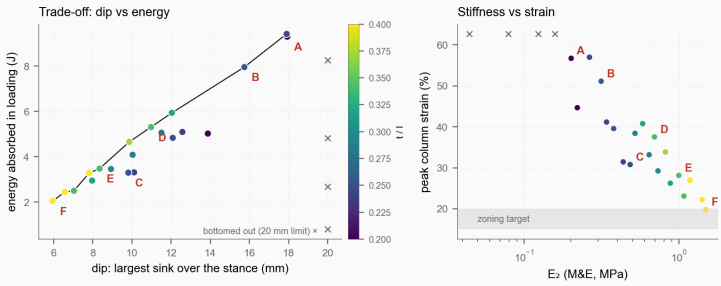


Fig. 3 Dip against energy for 24 cells: one near-linear front.

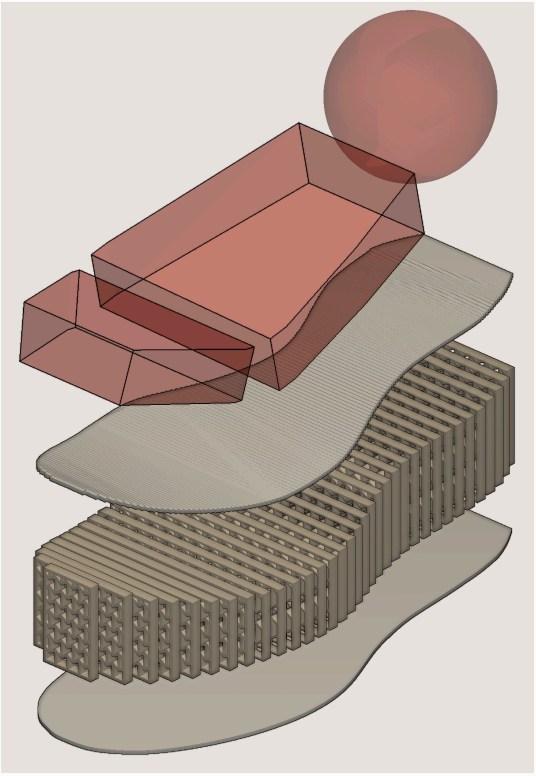


Fig. 4 The zoned midsole, exploded, under the pseudo-foot (red).

> *There is no free lunch: every extra millimetre of dip buys about 0.6 J of absorbed energy.*

- **Zoning wins.** A stiffer cell under the forefoot, where load concentrates; the softest, most auxetic cell under midfoot and heel.
- **ANSYS moves points, not the front.** Recalibrated to the 25-cell batch, a cell dips 3 % less to 13 % more; the trade-off holds.
- **Collapse is a feature.** Most cells reach the buckling plateau — where a re-entrant sole stays compact and caps the peak force.

STRAND 02

Chiral

Rings that turn together, ligaments that wind — and an airfoil that bends

The hexachiral lattice places a ring at every node of a triangular grid, joined to six neighbours by ligaments that leave it *tangentially*. Under compression every ring turns the same way and winds its ligaments up, so the lattice contracts in every direction at once. The ideal bending-only cell gives a cubic stiffness law and the limit ν = −1:

$$E = \frac{\sqrt{3}}{2} E_s \left(\frac{t}{L}\right)^3 \left(\frac{R}{r}\right)^2, \qquad \varphi = \frac{L}{2r}\,\varepsilon, \qquad \nu \to -1$$

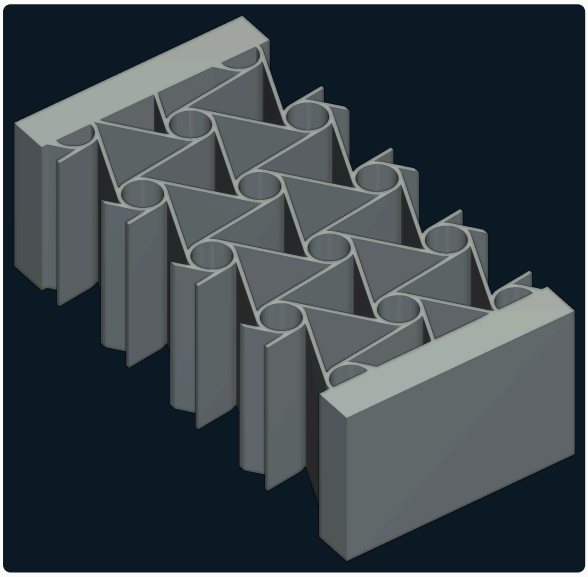


Fig. 5 The printed hexachiral sample, from its CAD.

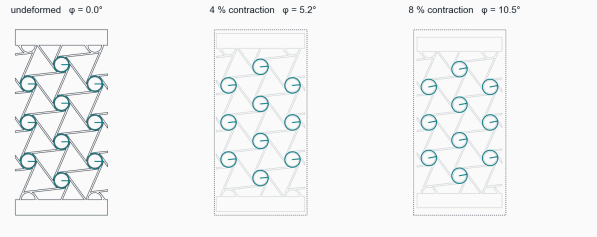


Fig. 6 Every ring turns by φ = (L/2r) ε as the lattice contracts (schematic).

Theory against a 3D coupon

Three closed forms were compared line by line — Spadoni & Ruzzene adds axial compliance, and two printed formulas (Prall & Lakes, Bornengo et al.) turned out to carry inconsistent prefactors. A finite 3D coupon (R = 25, t = 1 mm) was then compressed in seven ANSYS runs.

Model	E/E_s	ν
Bending-only limit	1.80×10^{-3}	−1
Spadoni & Ruzzene	1.74×10^{-3}	−0.954
ANSYS 3D coupon	2.89×10^{-3}	−0.819

The coupon is auxetic in every run, but stiffer and less auxetic than the ideal. Mesh refinement moves the modulus by at most 1.2 %, so the gap is physical — missing edge ligaments, end constraints, flexible rings — not numerical.

A morphing Eppler 420 trailing edge

The chiral mechanism spreads deformation smoothly through a structure — attractive for wings that change camber without hinges. A chiral core was mapped into a 330 mm chord Eppler 420 section and compared, in aluminium, with a straight-web core of almost the same mass (79.8 g vs 80.6 g), both driven by an internal actuator.

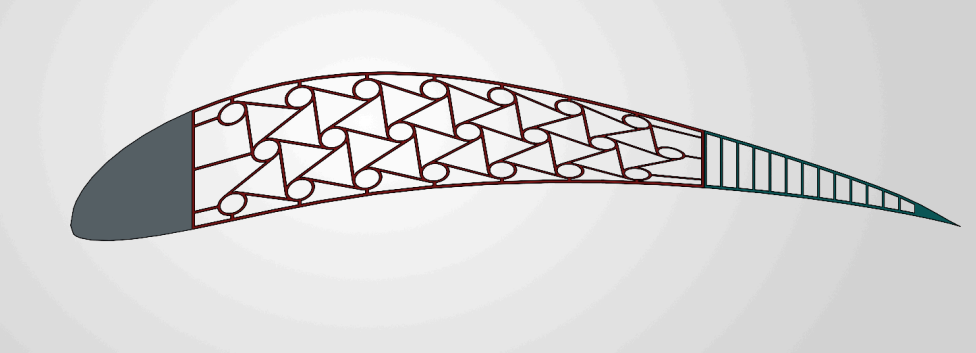


Fig. 7 The chiral-core section: solid nose, mapped chiral middle, cellular tail.

−21 %
actuator stroke for 0.5 mm of tip travel

2.5 ×
mechanical work vs straight web

0.61 mm
stroke at which the core reaches yield

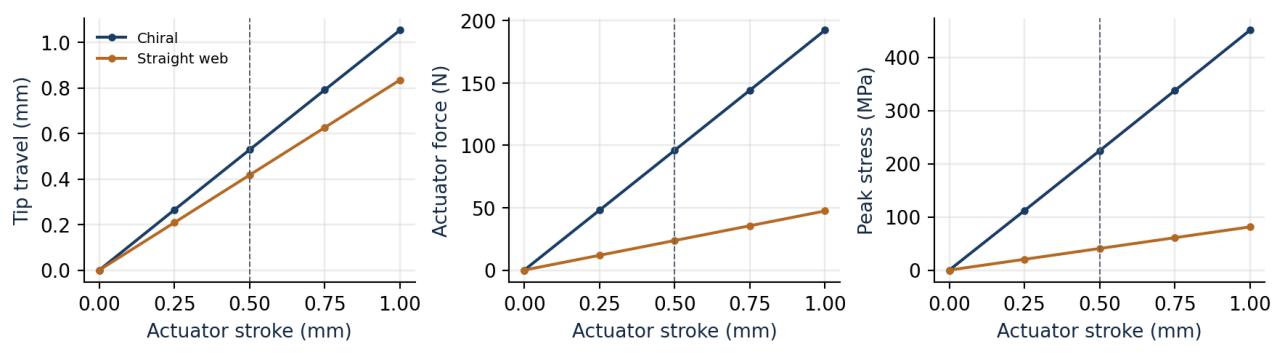


Fig. 8 Tip travel, actuator force and peak stress for both cores against stroke.

The chiral core turns **less stroke into more tip motion**, but pays in force and work — the classic morphing trade between a compliant mechanism's gain and the actuator and stress budget it demands.

STRAND 03

Rotating squares

Tiles that counter-rotate — exact theory, an independent solver, a soft specimen

Squares joined at their corners counter-rotate under load and open or close the gaps between them. A vector loop through one hinge gives the same pitch in both directions, so the lattice scales uniformly and ν = −1 exactly. Real hinges are short ligaments that also stretch; an energy method treats them as springs in series:

$$p(\theta) = a(\cos\theta + \sin\theta), \qquad \nu = -\frac{1}{1+8\rho}, \qquad \rho = \frac{w^2}{12\,b^2}$$

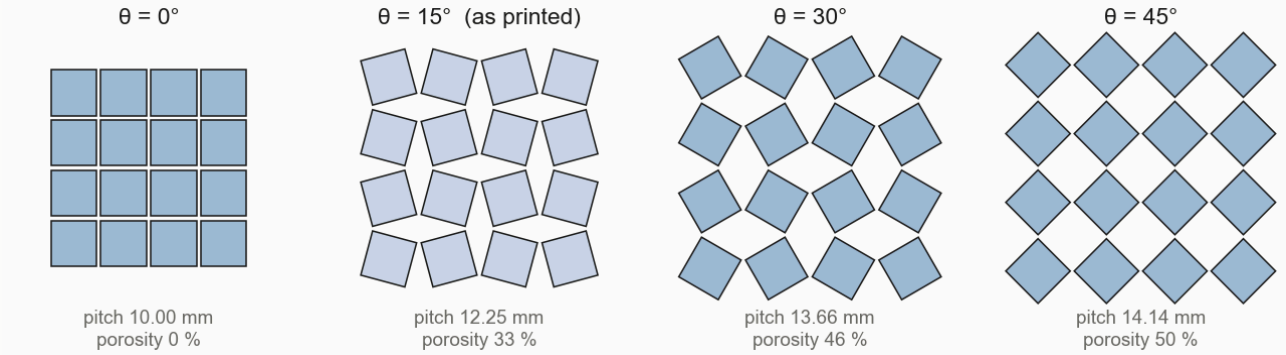


Fig. 9 The ideal lattice from closed to fully open; the open area grows from 0 to 50 %.

Three models that agree — and a limit that bites

For a stiff PLA design the energy method gives ν = −0.9915 and E_{eff} = 23.7 MPa; a beam finite-element solver written for the project reproduces both to 10^{-5}, and finite arrays differ by 0.1 %. The catch is strength: the ligaments yield below 1 % lattice strain, long before the kinematic locks or buckling.

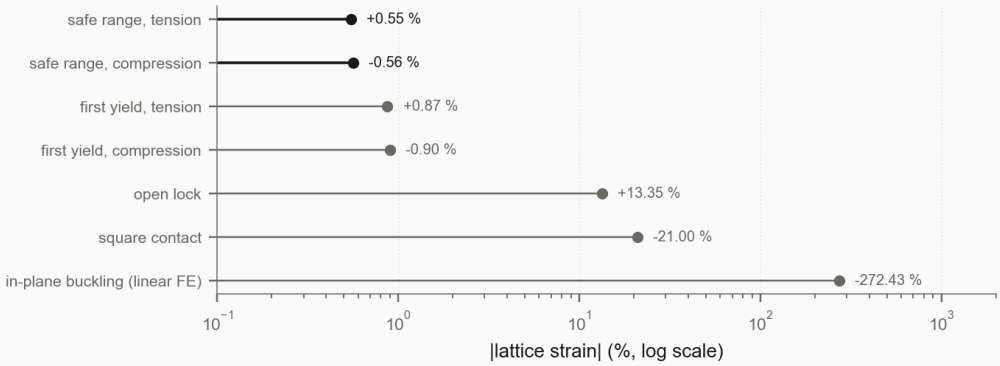


Fig. 10 Every limit on one log axis: yield governs by an order of magnitude.

A soft redesign in ANSYS

To let the mechanism move elastically, a TPU specimen was designed (36 squares, a = 8 mm, $\theta_0 = 30°$, $E_{eff} \approx 0.37$ MPa). An audit of its CAD found it 20 % smaller in plan and half as deep as the analytical design — a reminder to simulate the geometry that will actually be printed. Under 5 mm of compression in ANSYS it carried 29.9 N, and two internal gauges shortened together.

−9.6 %
axial gauge strain

−6.7 %
transverse gauge strain

−0.698
effective Poisson's ratio

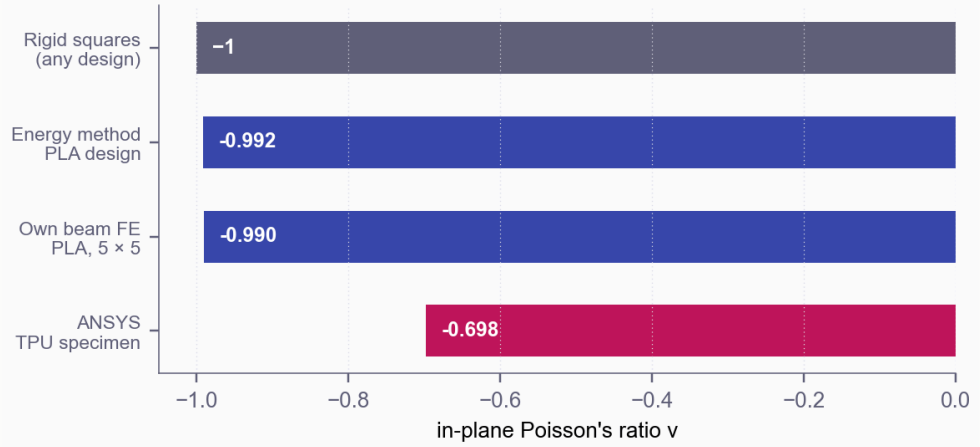


Fig. 11 Poisson's ratio across every description in the strand.

The specimen is clearly auxetic, but its transverse strain is only 70 % of the axial one. Stubby ligaments ($L_h/h \approx 1.5$) that stretch and shear, squares soft enough to deform, a six-row specimen held by solid end blocks, and nearly 10 % strain all pull it away from the rigid ideal. Rotating units remain the most compact route to isotropic auxeticity — the basis of expanding stents and hierarchical metamaterials.

STRAND 04

Tensegrity

Struts that never touch, held by cables that never slacken

A tensegrity holds its shape with compression struts suspended in a web of tension cables. Its stiffness comes from pre-tension — the *self-stress* σ — and the cable-to-strut stiffness ratio k. Following Gilewski et al. (2025), the strand studies 2D hexagonal tensegrity cells, whose continuum model has a soft mode in which one stiffness vanishes:

$$G_{12} = \frac{EA}{8ah}\left(\sqrt{3}\,k - 2\sigma\right) \;\to\; 0 \quad \text{at} \quad \sigma = \tfrac{\sqrt{3}}{2}\,k$$

Re-implemented, the model reproduces the paper's module properties to every printed digit. Rotating a cell by 60° cuts its vertical stiffness to a third and raises its Poisson's ratio tenfold — the strongest single design lever.

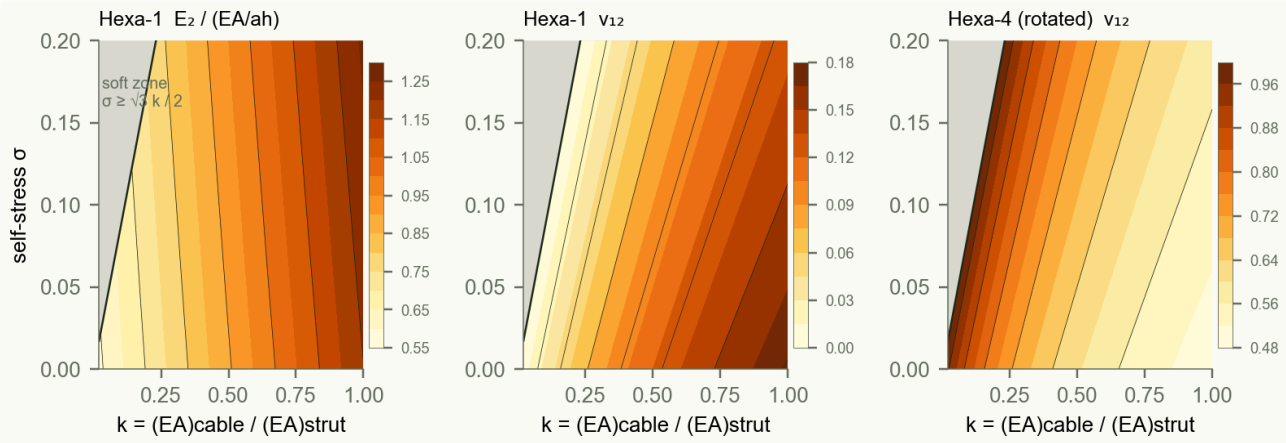


Fig. 12 Stiffness and Poisson's ratio over k and σ; grey is the soft zone.

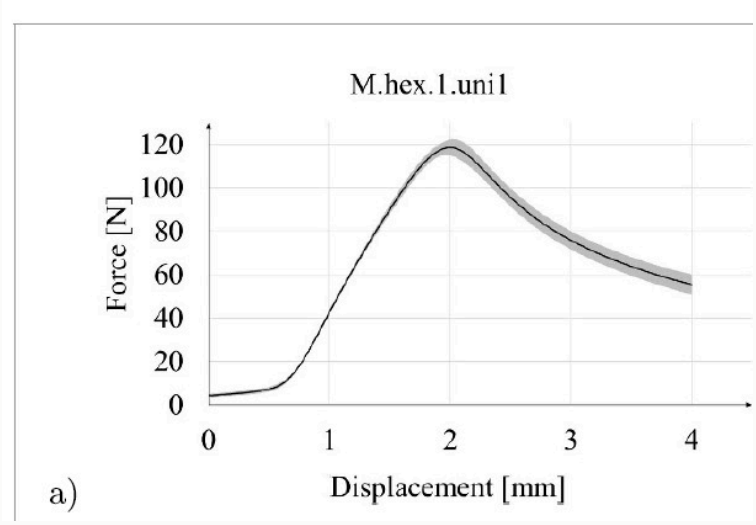


Fig. 13 The regular 2D lattice M.hex.1.uni1 under uniform load (mean of five specimens, ± one standard deviation): peak about 118 N near 2 mm, then softening as cells fail. From Gilewski et al. (2025).

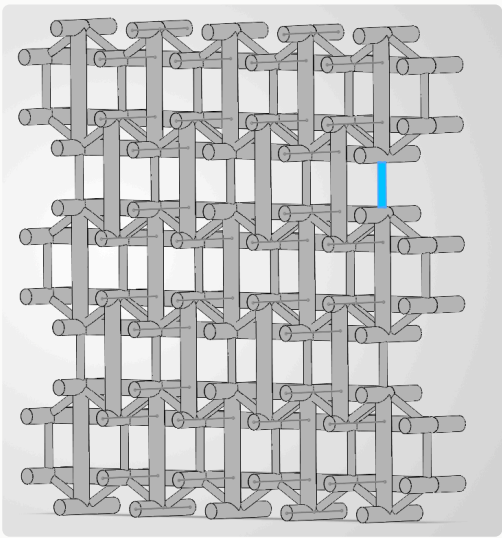


Fig. 14 The project's lattice in CAD: struts and cables in two planes.

From one sheet to a stacked 3D lattice

> *The aim of this strand is to compare a 3D lattice, made from layers of 2D lattices of hexagonal tensegrity cells, with the paper's single 2D lattice.*

The paper stops at planar sheets. Here the same hexagonal cells are stacked into a 3D column and analysed in ANSYS: fixed at the base, held by frictionless supports on 115 faces, and loaded from above. It deflects by at most **0.30 mm** (0.17 mm on average), and the deformation grows smoothly from support to loaded top. Unlike the single sheet — where a rotated layer forms a soft band that shears and fails first — the stacked layers brace one another, so no localised soft band appears.

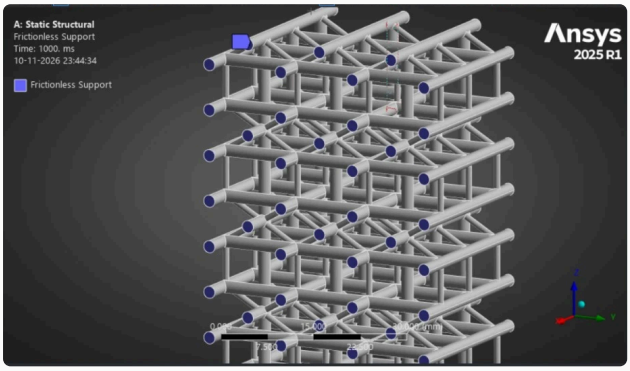


Fig. 15a Frictionless supports (blue) on 115 faces of the 3D column.

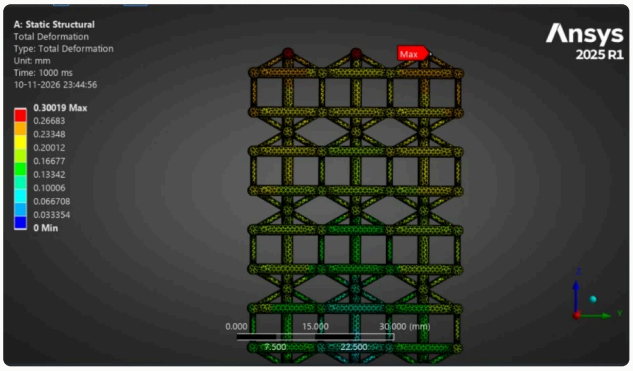


Fig. 15b Total deformation, maximum 0.30 mm at the loaded top.

Four lattices, one lesson

Strand	Ideal	Simulated	Biggest gap from theory
Re-entrant	ν −0.44 to −1	ν −0.14	thick joints resist folding
Chiral	ν −0.95 to −1	ν −0.82	finite coupon, flexible rings
Rotating squares	ν −0.99 to −1	ν −0.70	stubby ligaments, soft squares
Tensegrity	tunable via k, σ	stiff, smooth 3D response	rigid joints, no self-stress

Every printed lattice is a **weaker version of its ideal mechanism**. The closed-form models capture trends and rank designs, but printable walls, joints and ligaments are thick, so the real effect is smaller — from a modest shortfall in the chiral lattice to a seven-fold one in the re-entrant cell. Simulation is where that gap is measured; the applications — a cushioning midsole, a morphing wing, a soft expanding lattice, a graded load-limiter — are where it matters.

Full reports, code and data: github.com/ErrorNik14/metamaterials-and-applications